# SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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## Education

| Degree | Institute | CGPA | Duration |
|---|---|---|---|
| B.Tech., Computer Science and Engineering (AIML) | SRM Institute of Science and Technology, Chennai, Tamil Nadu | 8.98 / 10 | *2023 – 2027* |

## Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  *Software Engineering Intern* On-site
  - Built and deployed a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ REST APIs for business workflows.
  - Optimized **SQL queries, database indexing, and backend workflows**, reducing response time by **25%** during testing.
  - Integrated **Explainable AI** capabilities into the application to improve model transparency and support interpretable decision-making.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  *Google AI/ML Virtual Intern* Remote
  - Developed and evaluated **8+ ML models** across supervised and unsupervised learning tasks using 5+ datasets.
  - Applied **data preprocessing, feature engineering, EDA, and hyperparameter tuning** to improve model performance.
  - Evaluated real-world AI use cases and applied **bias mitigation** techniques to improve model reliability and responsible AI outcomes.

## Projects

- **Enterprise RAG & Agentic AI Assistant** *2026*
  *Technologies: Python, LangChain, LangGraph, Hugging Face, FAISS, Streamlit, SQL*
  - Built an enterprise knowledge assistant using **RAG, semantic search, and vector retrieval** over structured and unstructured documents.
  - Developed **agentic workflows** with tool calling, query routing, context optimization, and multi-step task execution using LangGraph.
  - Implemented evaluation pipelines measuring **answer relevance, retrieval quality, hallucination rate, latency, and response consistency**.

- **Demand Forecasting & Inventory Optimization** *2026*
  *Technologies: Python, Pandas, NumPy, Scikit-learn, TensorFlow, Streamlit*
  - Developed time-series forecasting models using **historical demand, seasonality, trends, and external signals** to predict future inventory requirements.
  - Compared statistical and machine learning approaches using **MAE, RMSE, and MAPE**, selecting models based on forecasting accuracy.
  - Built an interactive forecasting dashboard to visualize **demand trends, prediction intervals, inventory requirements, and model performance**.

- **AI Agent Orchestration & Workflow Automation Platform** *2026*
  *Technologies: Python, LangGraph, LangChain, FastAPI, Docker, PostgreSQL, Streamlit*
  - Built a multi-agent AI workflow that decomposes business tasks into **planning, retrieval, tool execution, and validation** stages.
  - Implemented **agent routing, tool calling, memory, structured outputs, and failure handling** using LangGraph and LangChain.
  - Developed FastAPI services and containerized the application with **Docker**, enabling modular deployment and scalable AI workflow execution.

## Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  *IEEE International Conference (SCEECS 2026)* View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, improving performance by **7% over baseline methods** and MCC by **8%**.

## Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

## Technical Skills

- **Programming**: Python, SQL, Java, C++
- **Data Science**: Pandas, NumPy, Scikit-learn, EDA, Data Preprocessing, Feature Engineering, Statistical Analysis
- **Generative AI**: LLMs, Prompt Engineering, RAG, AI Agents, Agentic Workflows, LangChain, LangGraph, Hugging Face
- **LLM Systems**: Vector Databases, FAISS, Semantic Search, Embeddings, Retrieval, Re-ranking, Context Optimization, LLM Evaluation
- **ML / Deep Learning**: Machine Learning, Deep Learning, NLP, PyTorch, TensorFlow, Transformers, BERT, RoBERTa
- **Data / Cloud / MLOps**: SQL, MySQL, MongoDB, AWS, Docker, Kubernetes, Git, GitHub, MLflow
- **Development**: FastAPI, Streamlit, REST APIs, Linux, Jupyter Notebook